

FOOD WEB OR FOOD CHAIN - FULL MIND MAP

Topic-wise | Hinglish | SSC / BPSC / BSSC / Railway

Topics in this chapter:

1. Food Chain - Order & Basic Structure
2. Producers, Consumers & Trophic Levels
3. Biotic/Abiotic Link & Decomposers
4. Pollinators & Food-web Effects
5. Primary & Secondary Productivity
6. Important Ecological Pair

1. Food Chain - Order & Basic Structure

- Grass, Grasshopper, Frog, Snake - kisi organism ko nutrition provide karne ke liye consumption kiye jane wale kisi bhi substance ko food kaha jata hai. plants svaposhi hote hain aur apna food svayan banate hain, jabki animal food ke liye plants par dependent rahate hain. [Q1]
- Grass, goat and human - ghas, bakari aur manushy ek food chain banate hain- ghas → bakari → manushy ↓ ↓ (svaposhi) (shakahari consumer) (sarvahari consumer) [Q2]
- nature ka sabse clear pahalua yeh hai ki energy ko ek living organism se dusare living organism tak jana hi padata hai. jab shakahari animal plants ko khate hain, to energy plants se animals mein sthanantarit hoti hai. kisi ecosystem mein kuchh animal dusre living organisms ko khate hain, jabki kuchh mrit organic substance par food karte hain. baad wale aparad food chain (Detritus food chain) banate hain. har ek chain mein aam taur par aise sirf char se panch relation hote hain. [Q5]
- Diatoms-Crustaceans-Herrings - ek common marine food chain ka sahi kram hai- dayatam → krasteshiyans → hering [Q17]

2. Producers, Consumers & Trophic Levels

- Primary Producer - food chain kisi fixed area ke andar organisms ke food kram ke is wajah se carbon aur energy ka pravah hai. organisms ko nutrition levels (trophic levels) mein vibhajit kiya jata hai, jo is bat par dependent karte hain ki koi vyaktigat organism apna food kaise obtained karta hai. [Q3]

EXAM TRAP primary consumer seedhe utpadakon par food karte hain; hiran clear roop se shakahari primary consumer hai. chintiyon bahut alag-alag group hain; species aur ahar ke ke mutabik ve plants substance ya madhuras kha sakti hain, dusre organisms ka shikar kar sakti hain, mrit substance kha sakti hain ya sarvahari ho sakti hain. isliye sabhi chintiyon ko ek saath primary consumer manana bahut zyada samanyikaran hai, yadyapi kuchh chinti species primary consumer ke trophic level par ho sakti hain. [Q6]

- primary as well as a secondary consumer - food chain mein trophic level hote hain, arthat svaposhi, primary consumer, secondary consumer aur tritiyak upabhokta. primary consumer shakahari hote hain (jo plants par food karte hain), secondary consumer chhote mansahari hote hain (jo primary consumers par food karte hain) aur tritiyak consumer bade mansahari hote hain. isliye food chain mein manushy primary aur secondary dono consumer hai, jo plants, utpadon aur mans dono par food karta hai. [Q8]

- Phytoplanktons - padapaplavak, jinhe sukshma algae bhi kaha jata hai, sthaliy plants ke saman hote hain kyunki inmein klorophil hota hai aur inhen living rahane aur increase ke liye suryaprakash ki zarurat hoti hai. zyadatar padapaplavak tairane wale hote hain aur ocean ke upari hissa mein tairate hain, jahan suryaprakash water mein enter karta hai. [Q9]
- Green Plants - plants svaposhi organism hote hain jo aam taur par light se energy (light-synthesis) ka use karke karbohidret, vasa aur protin jaise jatil organic compound banate hain. inhen svaposhi kaha jata hai aur ye sabhi ecosystems ki food chain ke liye mulabhut hain. [Q11]
- Green plants - food chain ka har ek charan ya level ek trophic level banata hai. trophic level-I mein svaposhi ya producer (arthat hare plants) shamil hote hain, jo solar energy ko stable karte hain aur ise paraposhiyon ya consumers ke liye available karate hain. [Q12]

3. Biotic/Abiotic Link & Decomposers

- Galat kathan: Air; biotic component ve living vastuen hain jo kisi ecosystem ko akar deti hain. biotic components mein aam taur par shamil hote hain: producer: jaise paudhe. [Q4]
- Decomposers - talab ke aquatic ecosystem mein kavak, baiktiriya aur phlajialets decomposer ke roop mein work karte hain. consumers ka pratinidhitv juplavak, svatantr roop se tairane wale aur tal par rahane wale rupon dwara kiya jata hai. [Q10]

EXAM TRAP producer abiotic substances aur solar energy ko biotic substance mein badalakar abiotic components ko living community se jodate hain. decomposer mrit organic substance ko vighatit kar inorganic nutrient tattvon ko soil, water aur air mein lautate hain, isliye ve living aur nonliving components ko viparit disha mein jodate hain. kyunki options mein 'producer aur decomposer dono' nahi diya gaya hai, maujuda vivadit (*) jawab uचित hai. [Q13]

- decomposer nature ke punarchakrak hote hain kyunki ve living praniyon ke mrit sharir mein paye jane wale organic substance ko todte hain. 'organic substance' term us substance ko sandarbhit karta hai jo living organisms se ata hai. [Q16]

4. Pollinators & Food-web Effects

- titaliyan anek pushpiy plants ke paragan mein important bhumika nibhati hain, isliye unaki ghatati number plants ke paragan par negative effect dal sakti hai. titaliyan food chain mein lower sadasy ke roop mein work karti hain. [Q7]

5. Primary & Secondary Productivity

- primary productivity woh dar hai jis par svaposhi/producer energy ko organic substance ya biomass mein badalate hain. secondary utpadakata woh dar hai jis par paraposhi consumer atmasat kiye gaye food se biomass banate hain. [Q14]

6. Important Ecological Pair

- Green plants - Ecosystem - sawal mein diye gaye sabhi pair ecosystem se related hain, lekin sirf (c) sahi sumelit hai. ecosystem living organisms ka ek community hai jo apne environment ke nonliving components ke saath sanyukt roop se present hota hai. chunki plants living organism hain, isliye ve ecosystem se related hain. [Q15]